

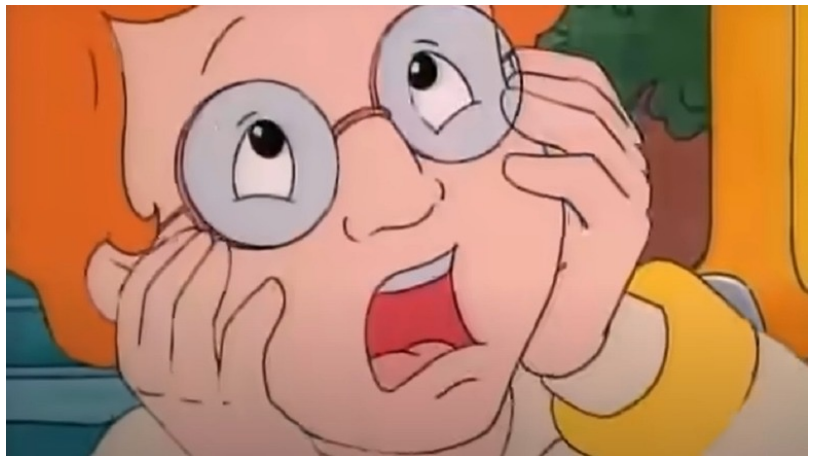
The M-CATs

Biology Chapter 9: The Digestive System

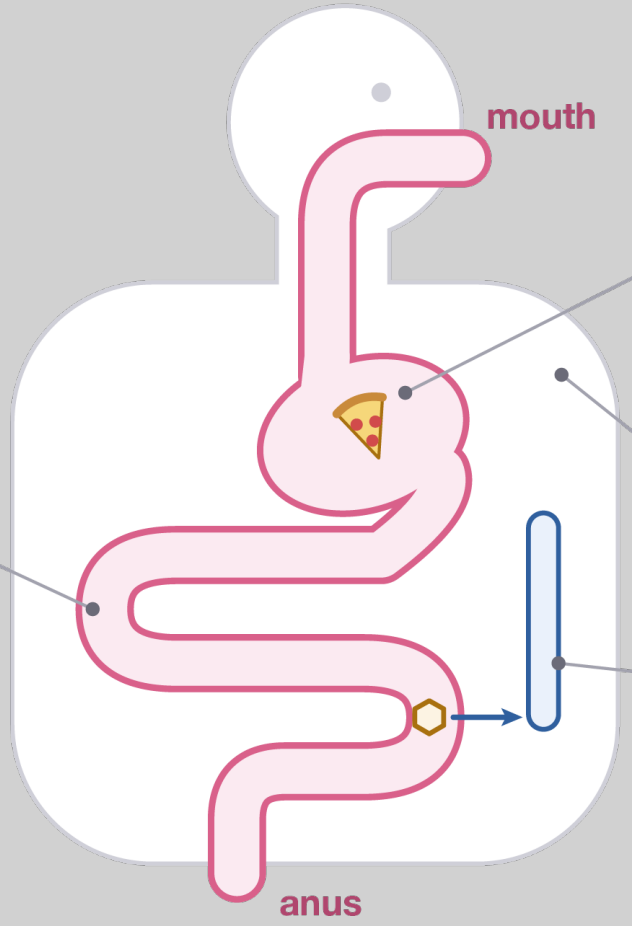
9.0



Did Ms Frizzle Really go **inside** Arnold?



outside the body



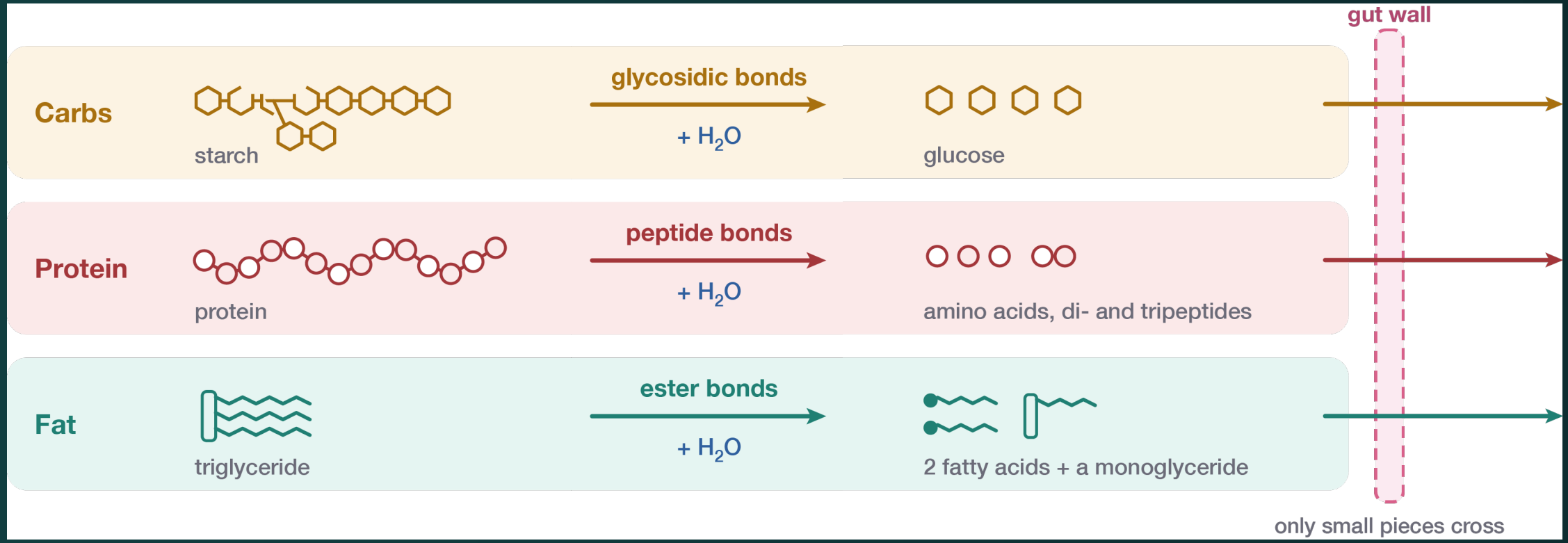
pizza in the stomach
still outside you

the body's tissue
this is inside you

lumen of the gut
open to the outside

absorbed into blood
now it's inside





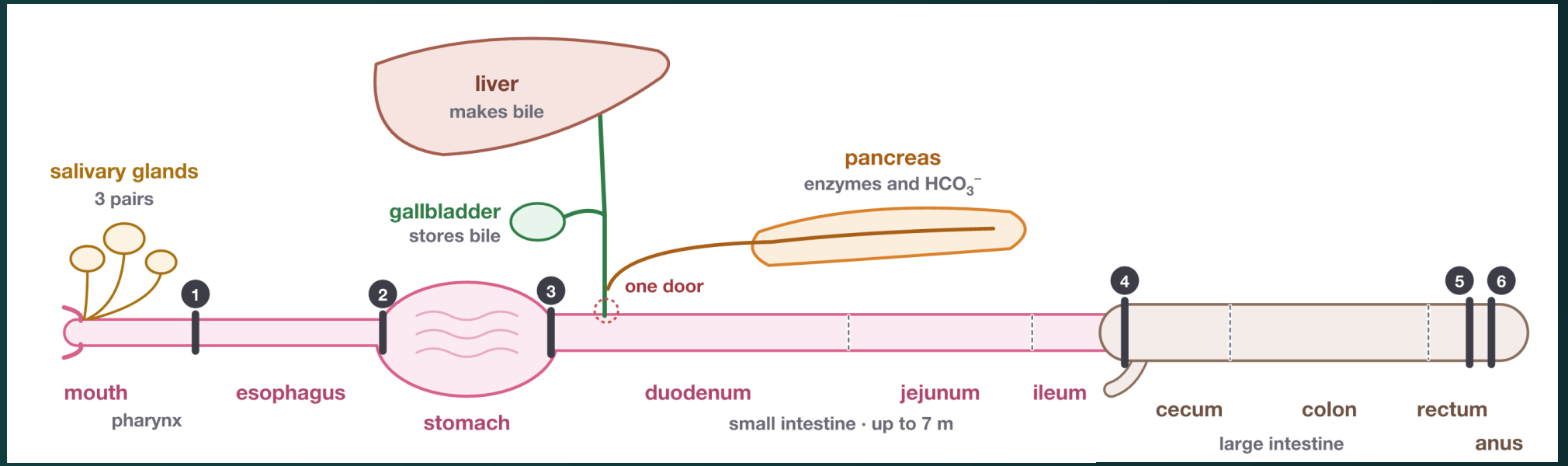
Extracellular vs Intracellular Digestion

Mechanical



Chemical





Break it down

Absorb it up

1

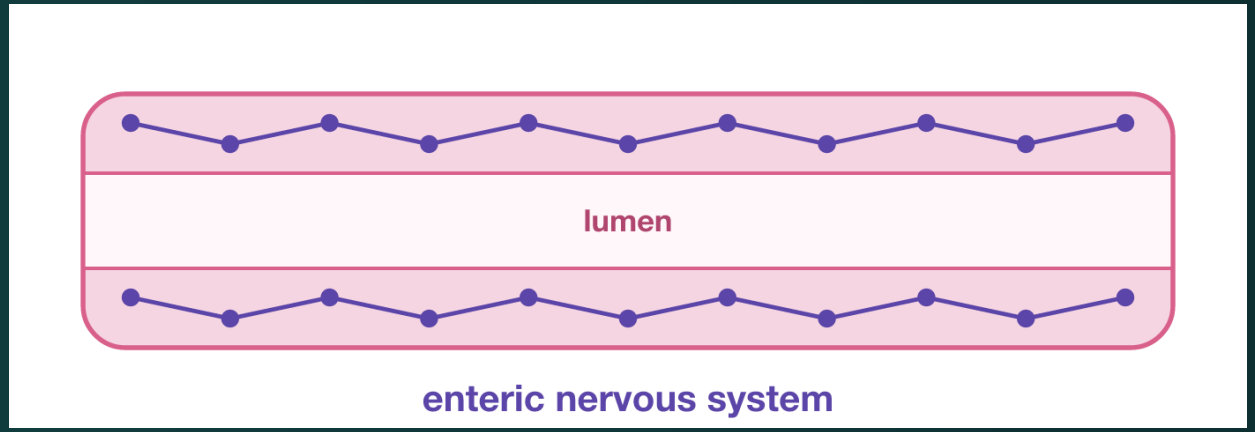
2

3

4

5

6



Parasympathetic

rest and digest

↑

more saliva and gut juices
more peristalsis

Sympathetic

fight or flight

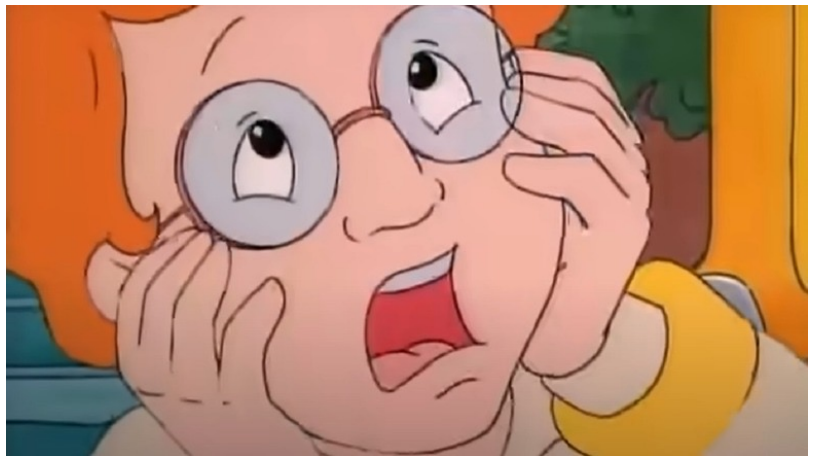
↓

less secretion
slower peristalsis
blood sent to the muscles

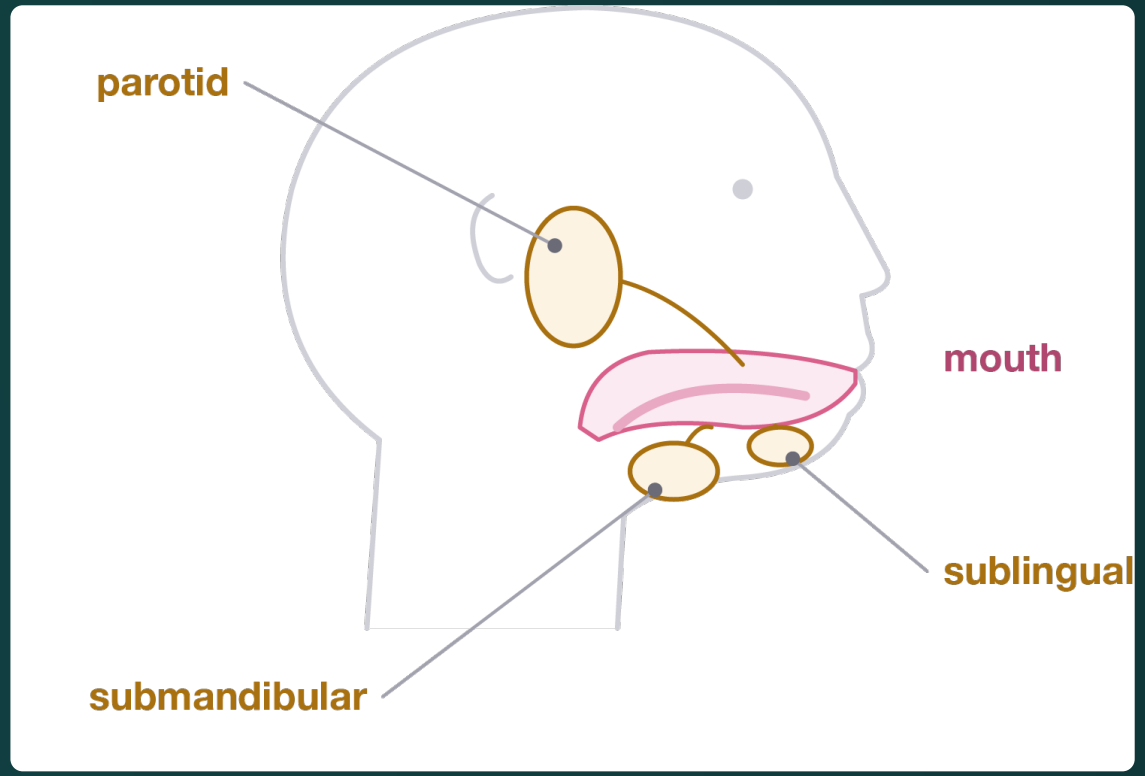


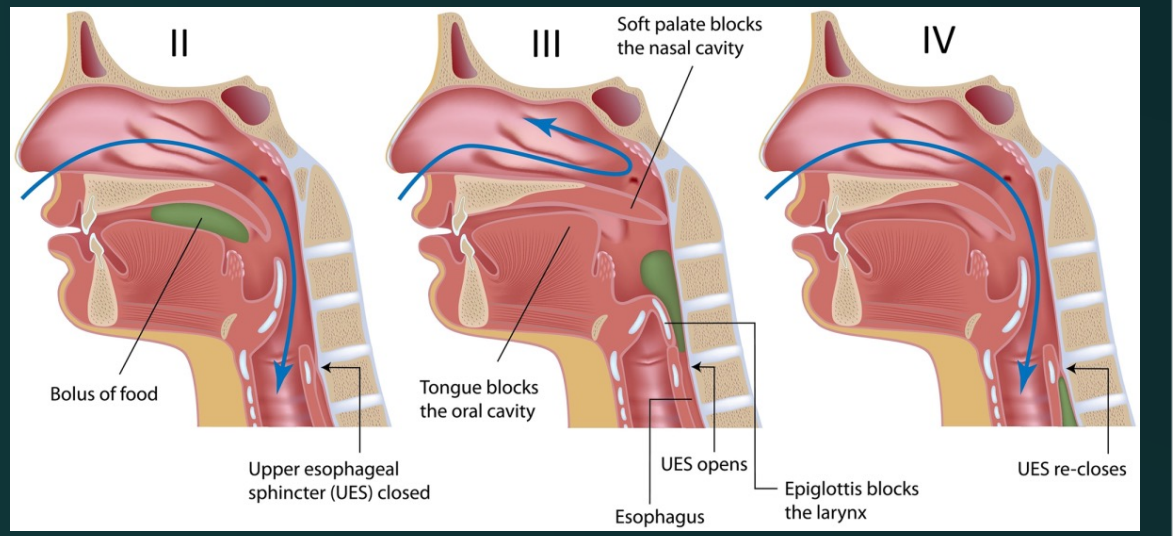
9.1

The journey into Arnold's Canal.



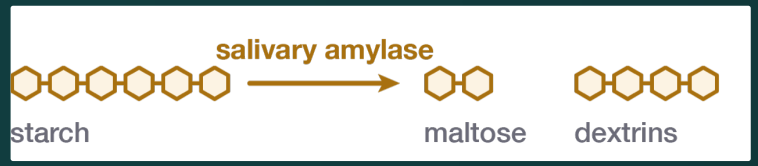






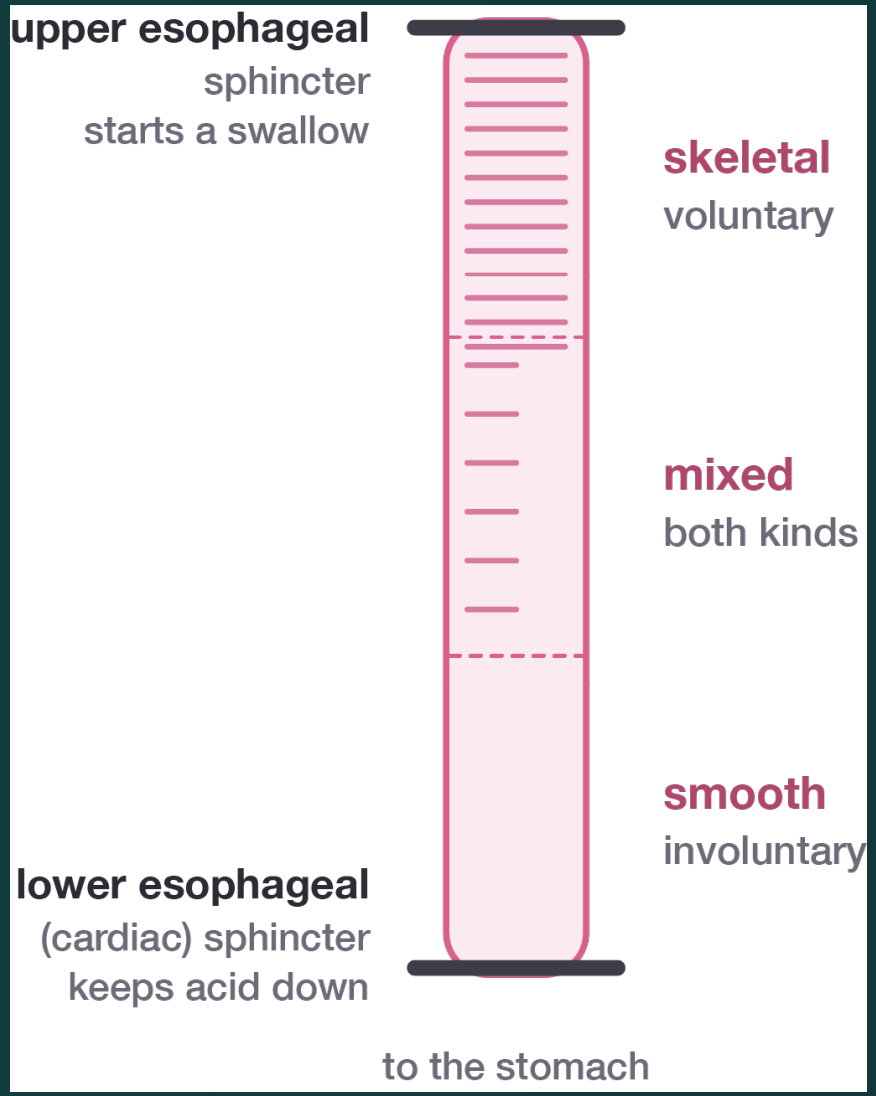
Chewing

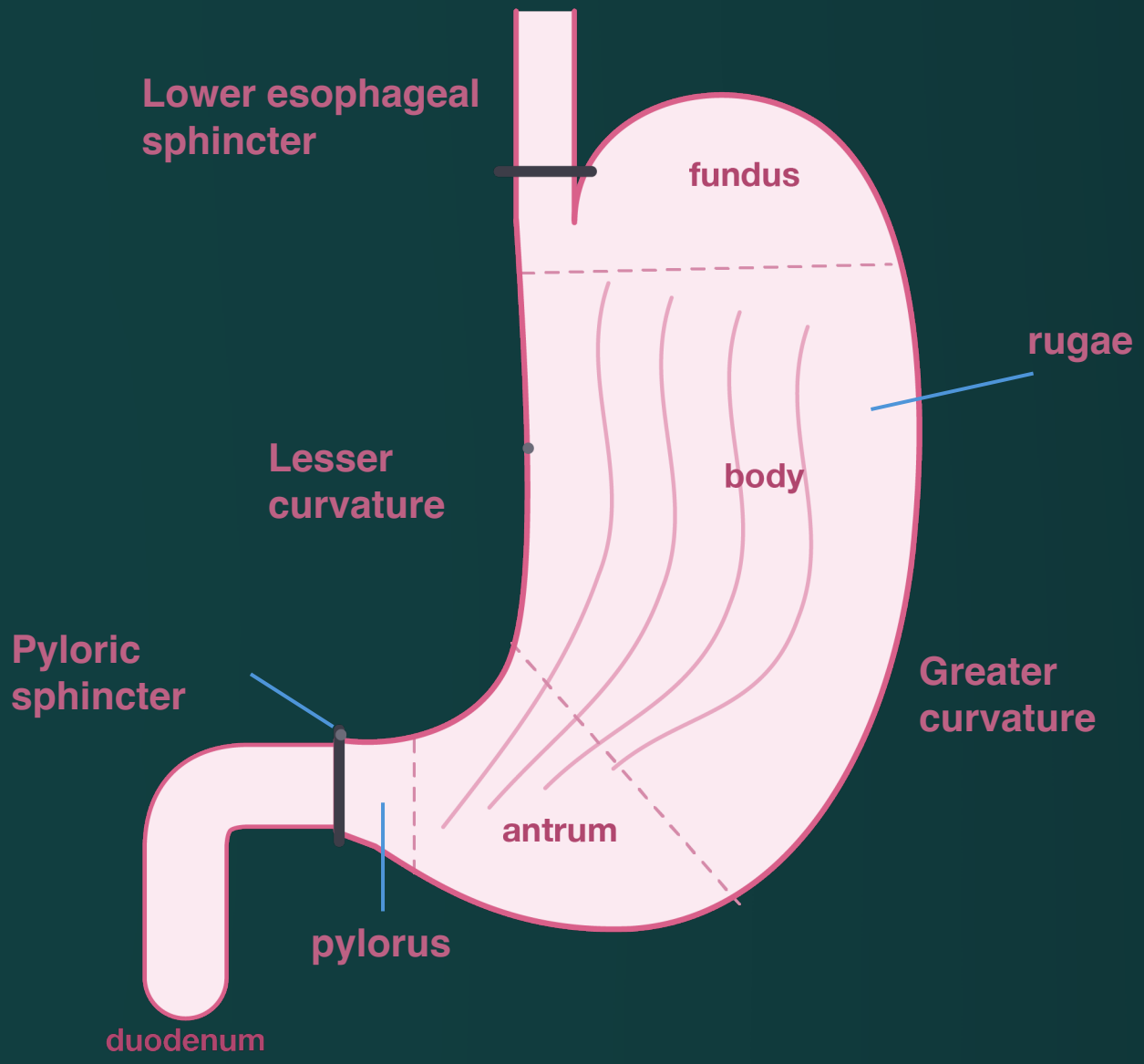
Salivary Amylase

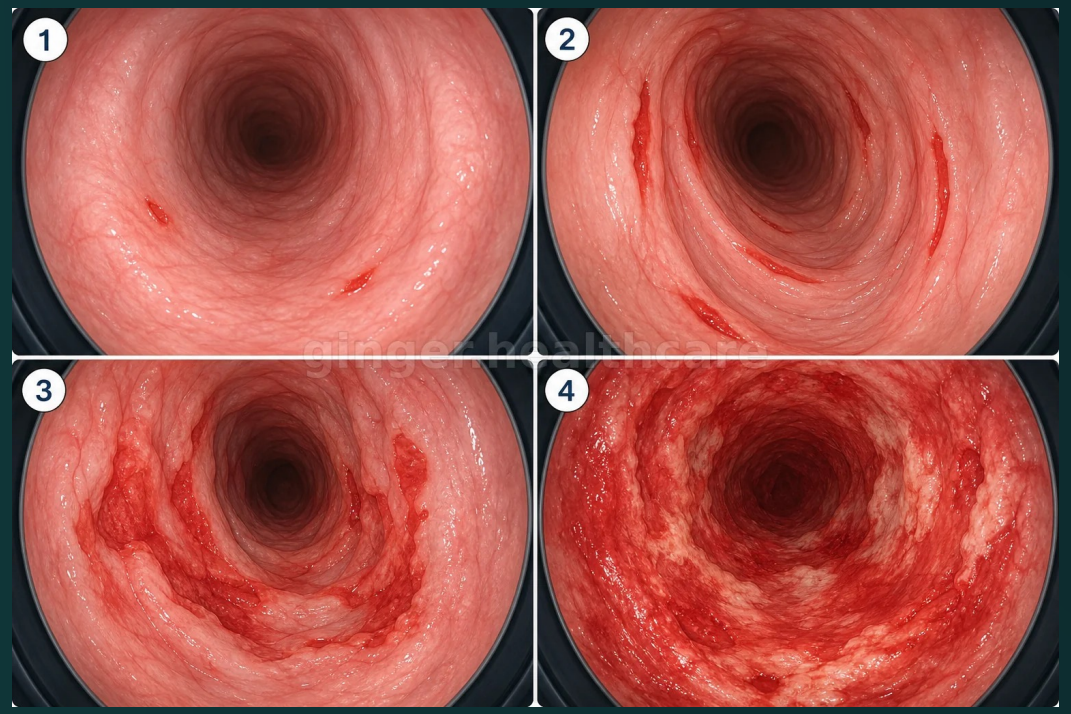


Lingual Lipase

The Esophagus

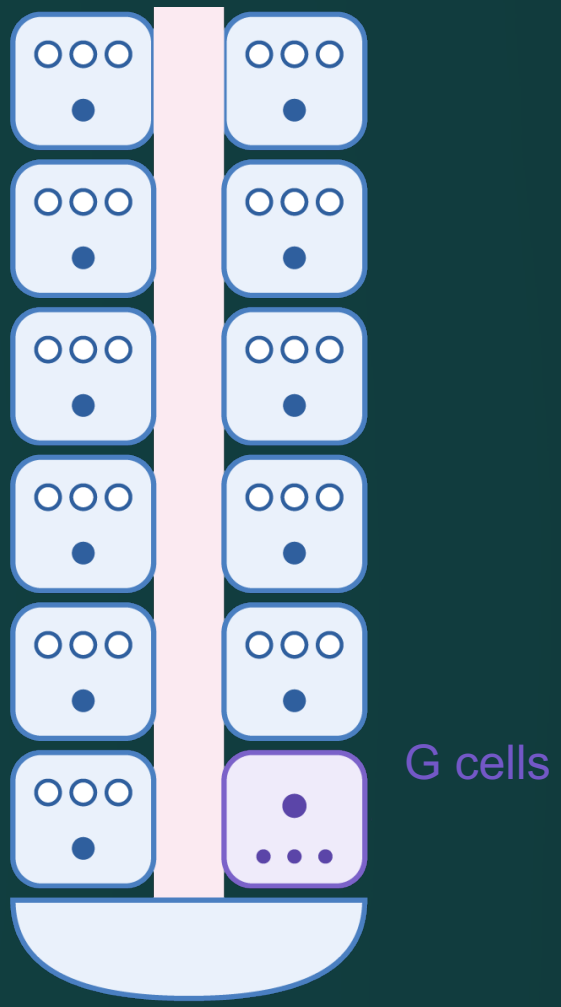






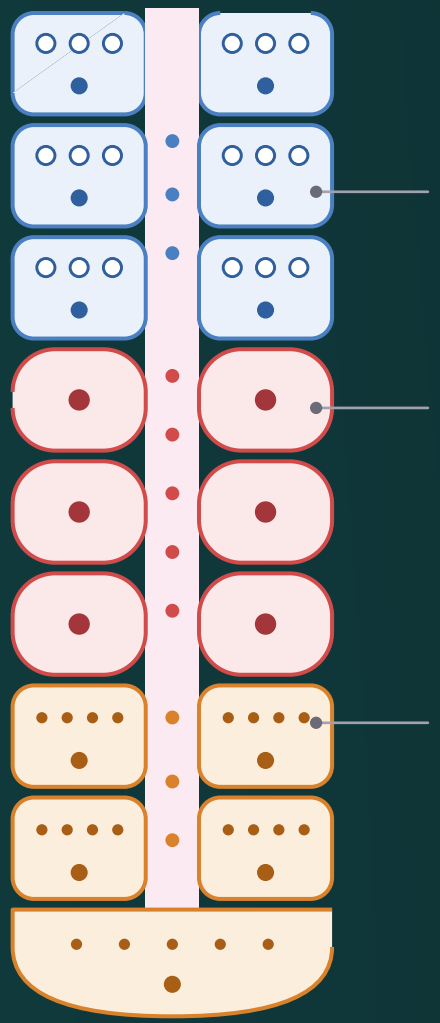


Sidenote: wtf is this ai image



Pyloric Gland

Antrum & pylorus

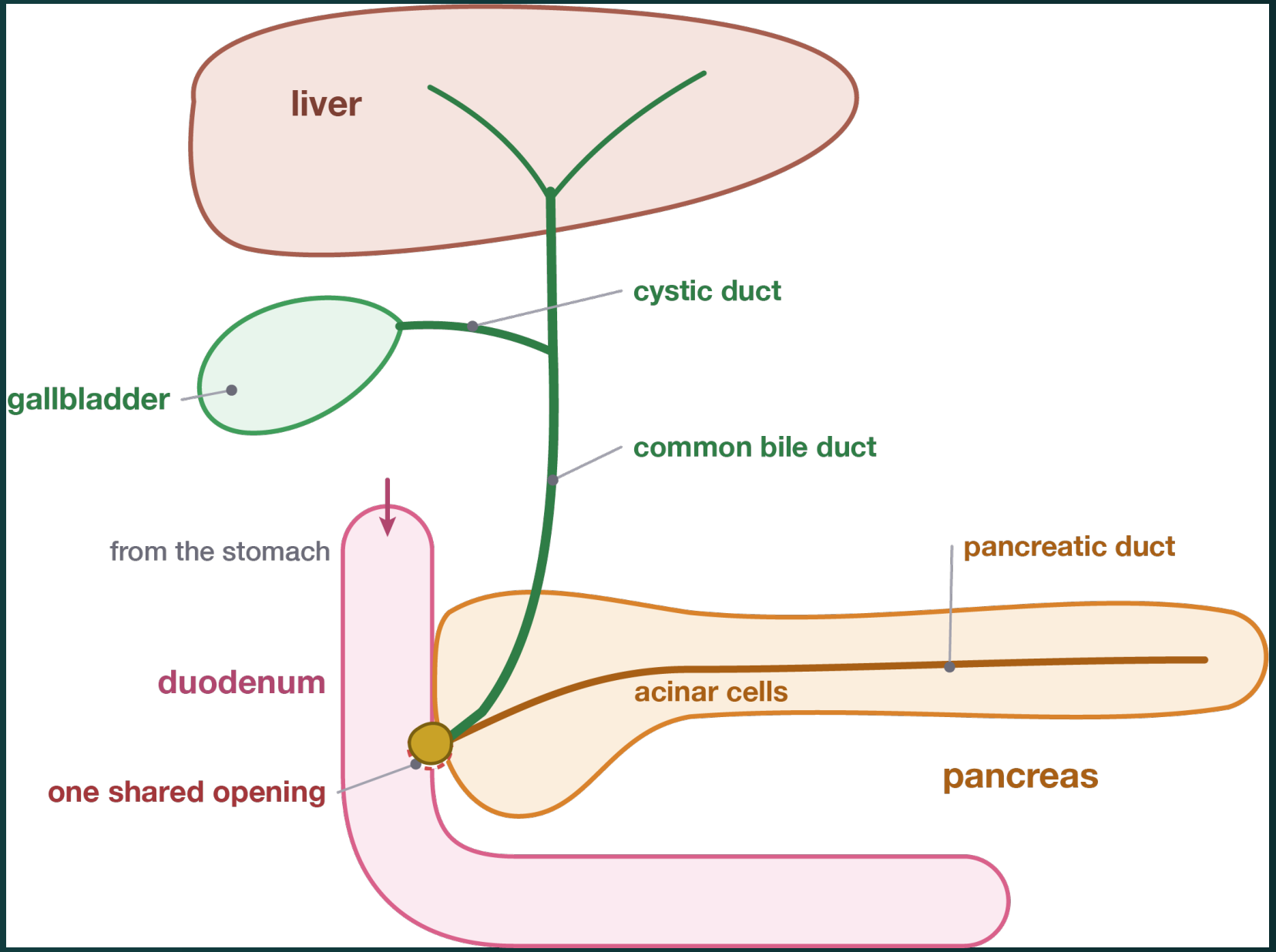


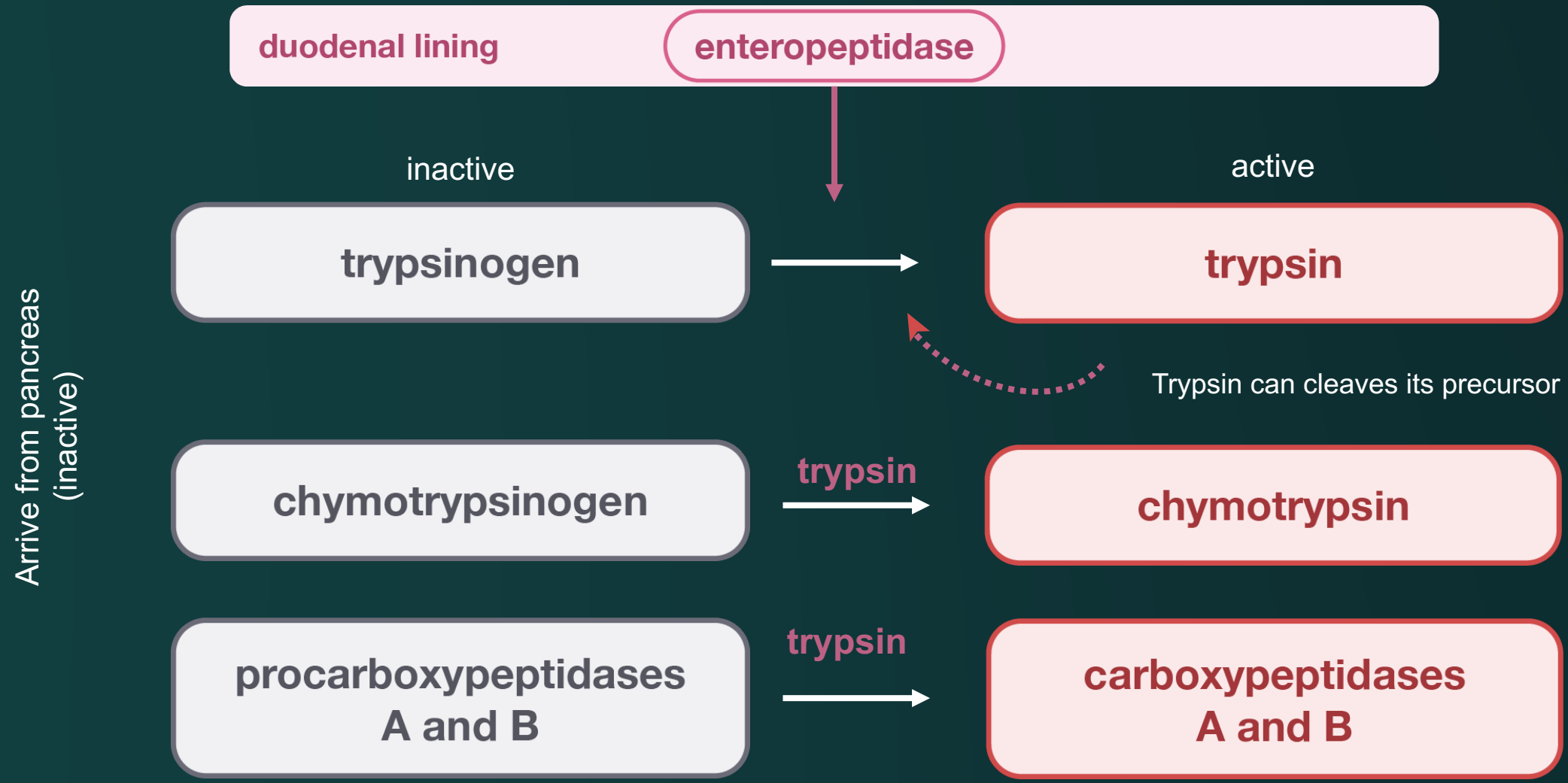
Gastric Gland

Fundus & body

Them chyme flows into the duodenum…


Secretin
acid enters the duodenum
S cells release secretin
pancreas sends HCO_3^-
stomach makes less HCl
stomach empties slower
acid neutralized
CCK
fat and protein enter
I cells release CCK
gallbladder squeezes out bile
pancreas sends enzymes
brain: you feel full
fat and protein digested

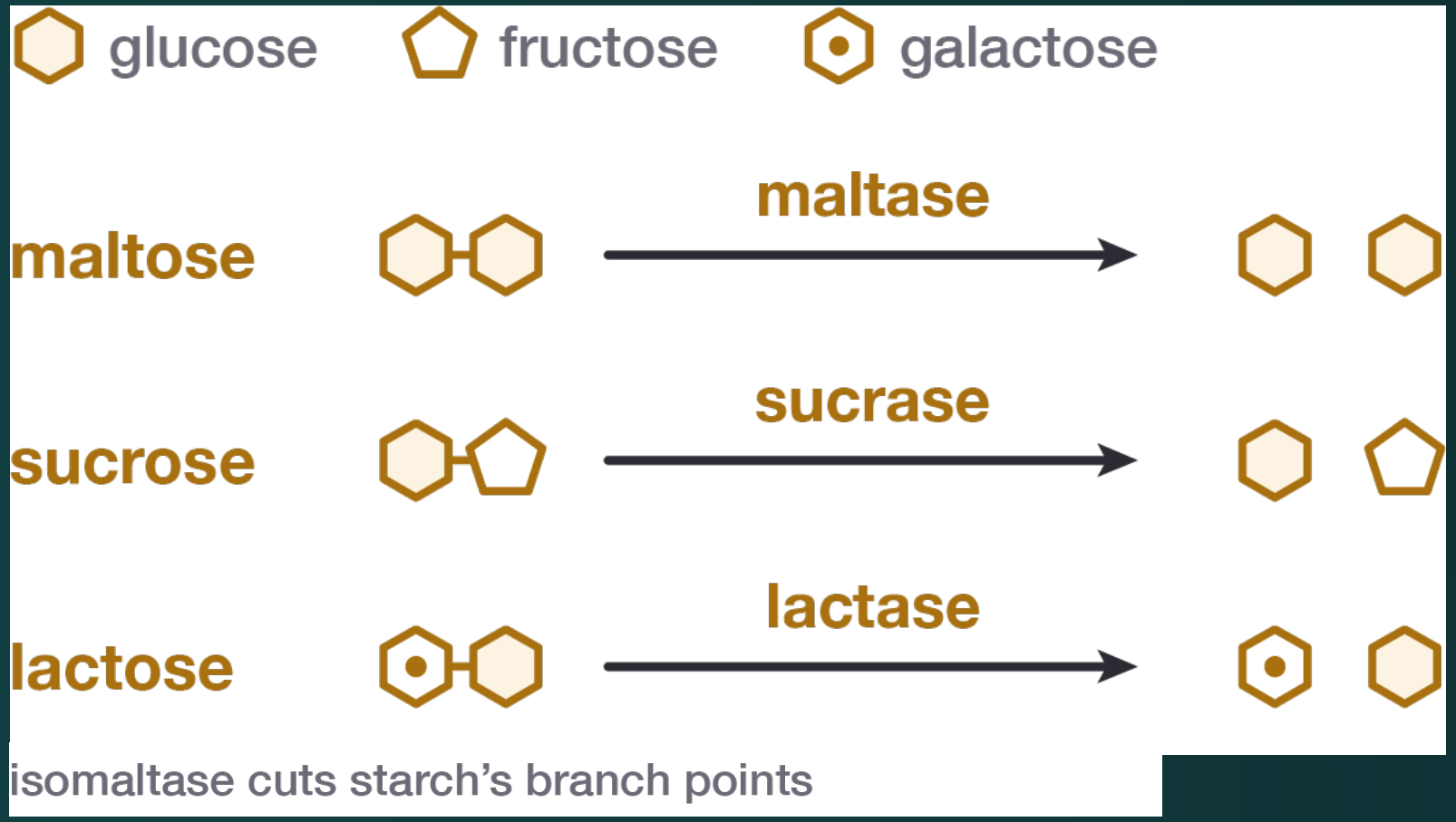

liver
cystic duct
gallbladder
common bile duct
from the stomach
pancreatic duct
duodenum
acinar cells
pancreas
one shared opening



So this breaks down big proteins into small peptides

Big sugars are broken down to disaccharides via pancreatic amylase

Finishing the Job





What about fats?

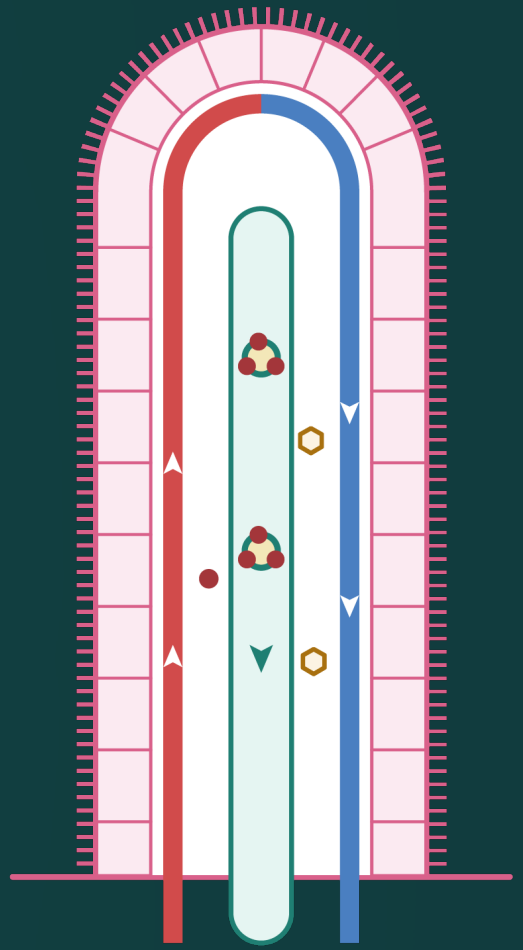


Bile?

Made up of:



Absorption and the liver



microvilli

Capillary?

Lacteal

Large Intestine & Rectum

